

JOEL SOFFO

EXPERIENCE	Machine Learning Scientist (Ph.D CIFRE Program) <i>AIRBUS</i>	Issy-les-Moulineaux, France July 2023 - July 2026 (<i>expected</i>)
	- Research on data-driven methods for solving physics-based problems. - Design and implementation of deep learning architectures for operator learning and Scientific ML. - Developed of dimensionality reduction techniques to accelerate high-dimensional simulation workflows.	
	Machine Learning Scientist <i>AIRBUS</i>	Issy-les-Moulineaux, France Sept. 2022 - July 2023
	- Conducted applied research in Scientific Machine Learning. - Implemented and benchmarked Neural Operators and PINNs for industrial simulation tasks.	
	Research Intern <i>AIRBUS</i>	Issy-les-Moulineaux, France Apr. 2022 - Sept. 2022
	- Evaluated state-of-the-art methods in Scientific Machine Learning (PINNs, Neural Operators). - Built experimental pipelines in PyTorch for PDE-based simulations.	
	Data Scientist Intern <i>LABELIANS</i>	Nemours, France Apr. 2021 - Sept. 2021
	- Optimized pricing models through statistical analysis. - Developed a user-facing analytics interface for marketing teams.	
EDUCATION	Ph.D in Applied Mathematics <i>Ecole Centrale de Nantes</i>	Nantes, France 2023 - 2026 (<i>expected</i>)
	- Advisor: Prof. Anthony Nouy - Research interests: Deep Learning, dimensionality reduction methods, Physics-Informed Machine Learning, Operator Learning.	
	Master Data Science <i>Institut Polytechnique de Paris (IP Paris)</i>	Palaiseau, France 2021 - 2022
PUBLICATIONS	A. Bensalah, A. Nouy, J. Soffo. Nonlinear manifold approximation using compositional polynomial networks. Accepted in <i>SIAM Journal of Scientific Computing</i>, 2026 M. Dus, G. Dusson, V. Ehrlacher, C. Guillot, J. Soffo. Comparison between tensor methods and neural networks for the computation of the Schrödinger equation ground state in electronic structure calculations. <i>ESAIM: Proceedings and Surveys</i>, 2024	

SELECTED PROJECTS	CEMRACS	2023
	Investigation of deep learning methods to solve Schrödinger equation	
	Generative Modeling for Financial Risk	2022
	- AI competition organized by BNP Paribas and Ecole Polytechnique - Developed stress-testing models using GANs and VAEs - Awarded Third Prize	
AWARDS AND HONORS	Ransomware Detection via Bitcoin Transactions	2022
	- Built feature extraction pipeline from blockchain data - Trained classification models to detect ransomware attack families	
	Third Prize, Generative Modeling Data Science Challenge by BNP Paribas / X	2022
	Best intern of the year, LABELIANS	2021
SKILLS	Academic Scholarship, Fondation Mathématique Jacques Hadamard (FMJH)	2020
	Languages: French (native), English (fluent)	
	Programming: Python, R, MATLAB	
	Tools: Git, VS Code, PyCharm	
TEACHING	Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, JAX	
	Teaching Assistant: Mathématiques pour les Sciences - First year undergraduates, <i>Sorbonne Université (Paris-VI)</i>	2023-2024
	Teaching Assistant: Numerical and Fourier Series - Second year undergraduates, <i>ESIEE Paris</i>	2023-2024
	Internship supervisor of Takumi HADA (ENSTA 2024) - Internship subject: Machine Learning for simulation workflows	
MANAGEMENT		